Experiment No: 8

Design a web page of your home town with an attractive background color, text color, an Image, font etc. (use internal CSS).

Date:

Competency and Practical Skills:

Relevant CO: 3

Objectives:

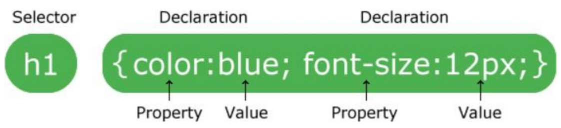
1. To understand how CSS works.

Theory:

Introduction To CSS

- CSS stands for Cascading Style Sheets
- CSS describes how HTML elements are to be displayed.
- CSS saves a lot of work. It can control the layout of multiple web pages all at once
- External stylesheets are stored in CSS files
- HTML was NEVER intended to contain tags for formatting a web page!. HTML was created to describe the content of a web page.
- When tags like <font>, and color attributes were added to the HTML 3.2 specification, it started a nightmare for web developers. Development of large websites, where fonts and color information were added to every single page, became a long and expensive process.
- To solve this problem, the World Wide Web Consortium (W3C) created CSS.
- CSS removed the style formatting from the HTML page!

CSS Syntax



- A CSS rule-set consists of a selector and a declaration block:
- The selector points to the HTML element you want to style.
- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS property name and a value, separated by a colon.
- declaration blocks are surrounded by curly braces.

Example: In this example all <p> elements will be center-aligned, with a red text color

Code	Output

<pre><!DOCTYPE html>
<html>
<head>
<style>
p {
  color: red;
  text-align: center;
}
</style>
</head>
<body>
 Hello World!
 These paragraphs are styled with CSS.
</body>
</html></pre>
	Hello World!
	
	These paragraphs are styled with CSS.

- p is a selector in CSS (it points to the HTML element you want to style: <p>).
- color is a property, and red is the property value
- text-align is a property, and center is the property value

CSS Selectors

- **CSS Element Selector**
  - The element selector selects HTML elements based on the element name.
  - Example:

```
p {
  text-align: center;
  color: red;
}
```

- The CSS id Selector
  - The id selector uses the id attribute of an HTML element to select a specific element.
  - The id of an element is unique within a page, so the id selector is used to select one unique element!
  - To select an element with a specific id, write a hash (#) character, followed by the id of the element.

- Example

```
<!DOCTYPE html>
<html>
<head>
<style>
#para1 {
  text-align: center;
  color: red;
}
</style>
</head>
<body>
<p id="para1">Hello World!</p>
<p>This paragraph is not affected
by the style.</p>
</body>
</html>
```

- CSS Class Selector
  - The class selector selects HTML elements with a specific class attribute.
  - To select elements with a specific class, write a period (.) character, followed by the class name.
  - Example
    - In this example all HTML elements with class="center" will be red and center-aligned:

```
<!DOCTYPE html>
<html>
<head>
<style>
.center {
  text-align: center;
  color: red;
}
</style>
</head>
<body>
<h1 class="center">Red and center-
aligned heading</h1>
<p class="center">Red and center-
aligned paragraph.</p>
</body>
</html>
```

- **CSS Universal Selector**

- The universal selector (*) selects all HTML elements on the page.
- Example

-

```
<!DOCTYPE html>
<html>
<head>
<style>
* {
  text-align: center;
  color: blue;
}
</style>
</head>
<body>
<h1>Hello world!</h1>
<p>Every element on the page will be
affected by the style.</p>
<p id="para1">Me too!</p>
<p>And me!</p>
</body>
</html>
```

- **CSS Grouping Selector**

  - The grouping selector selects all the HTML elements with the same style definitions.
  - To group selectors, separate each selector with a comma.
  - Example:

```
h1, h2, p {
  text-align: center;
  color: red;
}
```

- **The CSS Pseudo Class Selector**

  - Some selectors can be considered different because of the way the element they belong to works.
  - For example the anchor that creates a link between documents can have pseudo classes attached to it simply because it is not known at the time of writing the markup what the state will be.
  - It could be visited , not visited , or in the process of being selected.
  - CSS pseudo-classes are used to add special effects to some selectors. You do not need to use JavaScript or any other script to use those effects.
  - selector:pseudo-class {property: value}
  - CSS classes can also be used with pseudo-classes

- selector.class:pseudo-class {property: value}
- **Example**

```
a : link { color: red}
a : active { color: yellow}
a : visited { color: green}
a : hover { font-weight: bold}
a : link : hover {font-weight:bold}
```

- **Types Of CSS**
    - External CSS
    - Internal CSS
    - Inline CSS
- **Internal CSS**
    - An internal style sheet may be used if one single HTML page has a unique style.
    - The internal style is defined inside the <style> element, inside the head section.
    - Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
body {
  background-color: linen;
}
h1 {
  color: maroon;
  margin-left: 40px;
}
</style>
</head>
<body>
<h1>This is a heading</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

-CSS Background Color

- The background-color property specifies the background color of an element.
- With CSS, a color is most often specified by:
    - a valid color name - like "red"

- a HEX value - like "#ff0000"
- an RGB value - like "rgb(255,0,0)"

Example:

```
body {
  background-color: lightblue;
}
```

-CSS Text Color

- text color can be set using color property

Example:

```
<h1 style="color:Tomato;">Hello
World</h1>
```

Implementation:

Design a web page of your home town with an attractive background color, text color, an

```
<!DOCTYPE html>

<html lang="en">



<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Ahmedabad</title>

        <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css">



    <style>

        body::before {

            position: absolute;
```

Image, font etc. (use internal CSS).

```
    content:      "";
    margin-top: 0;
    margin-left:  0;
    height: 100vh;
    width: 100vw;
    background-color: aliceblue;
    opacity: 0.4;
    z-index: -1;
}

body {
    margin: 0;
    height: 100vh;
    width: 100vw;
    background: url("bg_img.webp");
    background-size: cover;
    background-repeat: no-repeat;
    position: relative;
    z-index: 1;
    overflow: hidden;
}

#navbar {
    height: 40px;
    width: 100%;
    display: flex;
```

```
    flex-direction: row;
    justify-content: flex-start;
    padding: 5px;
    background-color: rgb(135, 175, 209);
    align-items: center;
}

#navbar a {
    font-size: 18px;
    text-decoration: none;
    margin-right: 30px;
    color: #15163b;
    font-weight: bold;
    margin-left: 10px;
}
#navbar i{
    font-size: 20px;
    margin-left: 10px;
    margin-right: 10px;
}

h1 {

    text-align: center;
    color: #15163b;
}
```

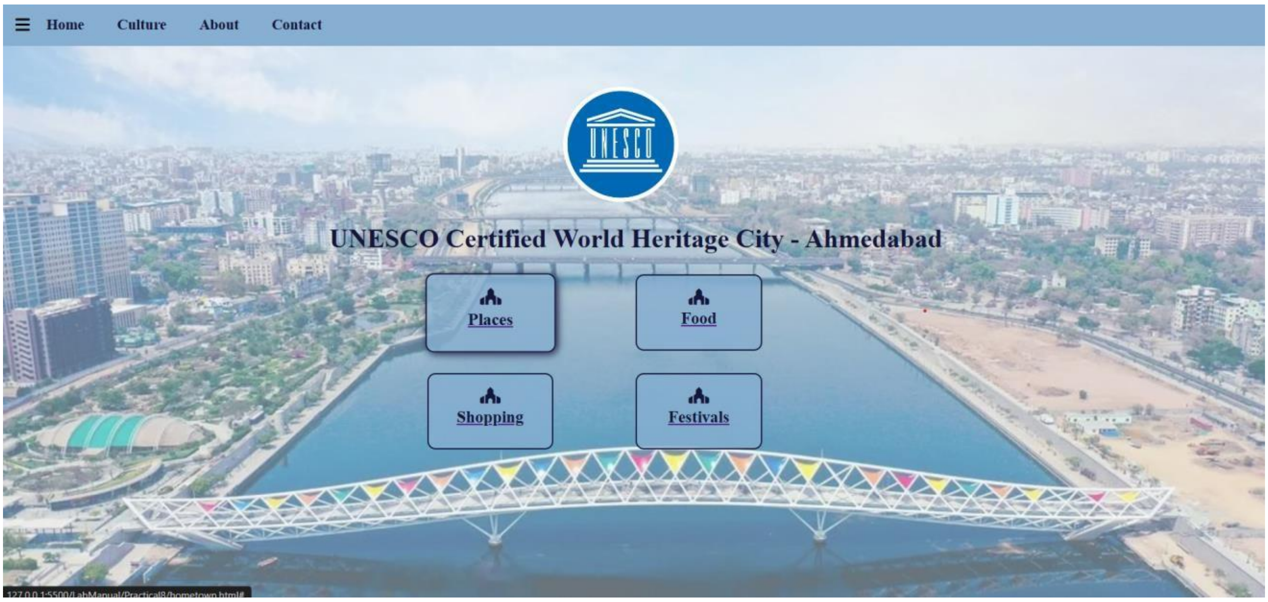
```
img {
    height: 140px;
    width: 140px;
    border-radius: 50%;
    margin-top: 50px;
    margin-left: 680px;
    object-fit: cover;
    transition: box-shadow 0.2s ease-in-out;
}

img:hover {
    box-shadow: 2px 2px 10px #15163b;
    transform: scale(1.03);
}

.content {
    width: 100%;
    height: 100px;
    display: flex;
    justify-content: center;
    align-items: center;
    margin-top: 20px;
}

.box {
```

```
        color: #15163b;
        height: 80px;
        width: 150px;
        background-color: rgba(135, 175, 209,0.8);
        border-radius: 10px;
        border: 2px solid #15163b;
        display: flex;
        flex-direction: column;
        justify-content: center;
        align-items: center;
        margin-right: 100px;
        font-weight: bold;
        font-size: 20px;
        padding-top: 10px;
        transition: 0.2s ease-in-out 0.1s;
    }
    .box:hover{
        transform: scale(1.03);
        box-shadow: 2px 2px 10px #15163b;
    }
    .box p{
        margin-top: 5px;
        text-decoration: none;
    }
  </style>
</head>
```

```html
<body>
    <nav id="navbar">
        <i class="fa-solid fa-bars"></i>
        <a href="#">Home</a>
        <a href="#">Culture</a>
        <a href="#">About</a>
        <a href="#">Contact</a>
    </nav>
    <img src="./unesco.jpg">
    <h1>UNESCO Certified World Heritage City - Ahmedabad</h1>
    <div class="content">
        <a href="#">
            <div class="box">
                <i class="fa-solid fa-place-of-worship"></i>
                <p>Places</p>
            </div>
        </a>
        <a href="#">
            <div class="box">
                <i class="fa-solid fa-place-of-worship"></i>
                <p>Food</p>
            </div>
        </a>
    </div>
    <div class="content">
```

```
      <a href="#">
        <div class="box">
          <i class="fa-solid fa-place-of-worship"></i>
          <p>Shopping</p>
        </div>
      </a>
      <a href="#">
        <div class="box">
          <i class="fa-solid fa-place-of-worship"></i>
          <p>Festivals</p>
        </div>
      </a>
    </div>
</body>

</html>
```

Output:


Home
Culture
About
Contact
UNESCO
UNESCO Certified World Heritage City - Ahmedabad
Places
Food
Shopping
Festivals

Conclusion:

This practical helped me design a web page using **internal CSS** where I learned to apply background colors, font styles, and images to make the webpage visually attractive. It improved my understanding of styling within a single HTML file.

Quiz:

1. Explain the syntax of the CSS.
2. What is internal CSS ?
3. Explain CSS class and Id selector.

Suggested Reference:

- https://www.w3schools.com/css/css_syntax.asp
- https://www.geeksforgeeks.org/types-of-css-cascading-style-sheet/

References used by the students:

Rubric wise marks obtained:

Rubrics	1	2	3	Total
Marks				

Experiment No: 9

Use Inline CSS to format your resume that you created in practical no 02.

Date:

Competency and Practical Skills:

Relevant CO: 3

Objectives:

1. To understand the use of Inline CSS.

Theory:

Internal CSS

- An inline style may be used to apply a unique style for a single element.
- To use inline styles, add the style attribute to the relevant element. The style attribute can contain any CSS property.

Example:

```
<!DOCTYPE html>
<html>
<body>
<h1 style="color:blue;text-align:center;">This is a heading</h1>
<p style="color:red;">This is a paragraph.</p>
</body>
</html>
```

Implementation:

Use Inline CSS to format your resume that you created in practical no 02.

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Vansh Hingrajiya - Resume</title>

  <link         href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet" />
```

```
 <style>
  body {
   font-family: 'Segoe UI', sans-serif;
   background-color: #f7f9fc;
   color: #333;
  }
  .section-title {
   color: #3d405b;
   border-bottom: 2px solid #3d405b;
   margin-bottom: 1rem;
  }
  a {
   color: #0044cc;
   text-decoration: none;
  }
  a:hover {
   text-decoration: underline;
  }
  .list-group-item {
   background-color: #f0f4ff;
  }
 </style>
</head>
<body>
 <div class="container py-4">
  <div class="text-center mb-4">
   <h1 class="fw-bold">Vansh Hingrajiya</h1>
   <p class="lead">Information Technology Undergraduate | Vishwakarma Government
```

```
Engineering College</p>

  </div>

  <div class="mb-4">

    <h3 class="section-title">Personal Information</h3>

    <p><strong>Address:</strong> E-505, Angan Residency, Nr. Dipak School, Nikol, Ahmedabad</p>

    <p><strong>Phone:</strong> <a href="tel:+919898441572">9898441572</a></p>

    <p><strong>Email:</strong> <a href="mailto:vanshhingrajiya7@gmail.com">vanshhingrajiya7@gmail.com</a></p>

    <p><strong>Date of Birth:</strong> 20-07-2006</p>

    <p><strong>Place of Birth:</strong> Ahmedabad</p>

    <p><strong>Nationality:</strong> Indian</p>

  </div>

  <div class="mb-4">

    <h3 class="section-title">Profile Summary</h3>

    <p>

      I am an IT student at VGEC, currently in my 4th semester with a CGPA of 7.6. I have a strong academic background with 93.69 percentile in HSC and 99.27 in SSC. My skills include C, Java, SQL, and beginner-level React Native with TypeScript. I'm actively enhancing my technical skills and have earned 538 Hackos on HackerRank.

    </p>

  </div>

  <div class="mb-4">

    <h3 class="section-title">Educational Information</h3>

    <ul class="list-group">

      <li class="list-group-item">

        <strong>Vishwakarma Government Engineering College</strong> <br />

        B.E. in IT (2023-2027) | CGPA: 7.6 <br />

        Core Subjects: C, Java, DSA, DBMS, Web Devlopment, OS, OOP

      </li>
```

```
    <li class="list-group-item">
     <strong>Shahjanand Vidhya Sankul</strong> <br />
     HSC (2021-2023) | Percentile: 93.69 | Percentage: 76.3%
    </li>
    <li class="list-group-item">
     <strong>Shree Chanakya Vidhya Sankul</strong> <br />
     SSC (2019-2021) | Percentile: 99.27 | Percentage: 94%
    </li>
   </ul>
  </div>
  <div class="mb-4">
   <h3 class="section-title">Professional Skills</h3>
   <ul class="list-unstyled">
    <li>✔ C Programming — <strong>Skillful</strong></li>
    <li>✔ Java Programming — <strong>Skillful</strong></li>
    <li>✔ SQL & Database Management — <strong>Skillful</strong></li>
    <li>✔ HTML, CSS & Bootstrap — <strong>Intermediate</strong></li>
    <li>✔ React Native with TypeScript — <strong>Beginner</strong></li>
   </ul>
  </div>
  <div class="mb-4">
   <h3 class="section-title">Experience / Achievements</h3>
   <p>
    <strong>Indus Hackathon 2025</strong> <br />
    Participated in Indus Hackathon held on 31st January 2025 at Indus University, Ahmedabad. Engaged in solving real-world problems, collaborated with peers, and developed time-bound tech solutions.
   </p>
```

```
  </div>

  <div class="mb-4">

   <h3 class="section-title">Hobbies</h3>

   <ul>

    <li>Playing chess and Sudoku</li>

    <li>Driving</li>

    <li>Cooking</li>

   </ul>

  </div>

 </div>

</body>

</html>
```

Output:

Vansh Hingrajiya

Information Technology Undergraduate | Vishwakarma Government Engineering College

Personal Information

Address: E-505, Angan Residency, Nr. Dipak School, Nikol, Ahmedabad

Phone: 9898441572

Email: vanshhingrajiya7@gmail.com

Date of Birth: 20-07-2006

Place of Birth: Ahmedabad

Nationality: Indian

Profile Summary

I am an IT student at VGEC, currently in my 4th semester with a CGPA of 7.6. I have a strong academic background with 93.69 percentile in HSC and 99.27 in SSC. My skills include C, Java, SQL, and beginner-level React Native with TypeScript. I'm actively enhancing my technical skills and have earned 538 Hackos on HackerRank.

Educational Information

Vishwakarma Government Engineering College
B.E. in IT (2023-2027) | CGPA: 7.6
Core Subjects: C, Java, DSA, DBMS, Web Devlopment, OS, OOP

Shahjanand Vidhya Sankul
HSC (2021-2023) | Percentile: 93.69 | Percentage: 76.3%

Shree Chanakya Vidhya Sankul
SSC (2019-2021) | Percentile: 99.27 | Percentage: 94%

Professional Skills

✔ C Programming — **Skillful**
✔ Java Programming — **Skillful**
✔ SQL & Database Management — **Skillful**
✔ HTML, CSS & Bootstrap — **Intermediate**
✔ React Native with TypeScript — **Beginner**

Experience / Achievements

Indus Hackathon 2025
Participated in Indus Hackathon held on 31st January 2025 at Indus University, Ahmedabad. Engaged in solving real-world problems, collaborated with peers, and developed time-bound tech solutions.

Hobbies

- Playing chess and Sudoku
- Driving
- Cooking

Conclusion:

By formatting my resume with **inline CSS**, I learned how to style individual HTML elements directly. This gave me a clear idea of element-specific customization though it is less reusable compared to other CSS methods.

Quiz:

1. Explain Internal CSS VS Inline CSS.
2. CSS stands for ______.
3. Which HTML tag is used to define an internal style sheet?

Suggested Reference:

- https://www.geeksforgeeks.org/types-of-css-cascading-style-sheet/

References used by the students:

Rubric wise marks obtained:

Rubrics	1	2	3	Total
Marks				

Experiment No: 10

Use External, Internal, Inline CSS to format Information Technology Department Web Pages that you created in Practical No.04

Date:

Competency and Practical Skills:

Relevant CO: 3

Objectives:

1. To understand use of External CSS

Theory:

External CSS

- An external file is a good idea when you have a number of pages, or even a complete site, which you need to control in terms of presentation.
- it saves lots of effort as at one time you would have needed to alter each page individually.
- With an external style sheet, you can change the look of an entire website by changing just one file!
- Each HTML page must include a reference to the external style sheet file inside the <link> element, inside the head section.
- External CSS file must be saved with a .css extension.
- **Example**

HTML Code : index.html

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet"
type="text/css" href="mystyle.css">
</head>
<body>
<h1>This is a heading</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

CSS Code : mystyle.css

```
body {
  background-color: lightblue;
}
h1 {
  color: navy;
  margin-left: 20px;
}
```

Implementation:

Use External, Internal, Inline CSS to format Information Technology Department Web Pages that you created in Practical No.04

HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>VGEC - Department of Information Technology</title>
    <link rel="stylesheet" href="style.css">
    <style>
        body {
            font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;
            margin: 0;
            line-height: 1.6;
            background-color: #f9fafc;
        }
        header {
            background-color: #002147;
            color: white;
            padding: 20px;
```

```
    text-align: center;
}
nav {
    background-color: #00509e;
    padding: 12px;
    text-align: center;
}
nav a {
    color: white;
    text-decoration: none;
    margin: 20px;
    font-weight: 600;
    transition: color 0.3s;
}
nav a:hover {
    color: #ffcc00;
}
section {
    padding: 25px 50px;
}
.about, .facilities, .faculty, .vision-mission, .contact {
    background: white;
    margin: 20px auto;
    border-radius: 10px;
    box-shadow: 0px 2px 8px rgba(0,0,0,0.1);
    max-width: 1000px;
```

```
    }
    footer {
      background-color: #002147;
      color: white;
      text-align: center;
      padding: 15px;
      font-size: 14px;
      margin-top: 40px;
    }
    table {
      width: 100%;
      border-collapse: collapse;
      margin: 15px 0;
    }
    table, th, td {
      border: 1px solid #ccc;
    }
    th {
      background-color: #00509e;
      color: white;
      padding: 10px;
    }
    td {
      padding: 10px;
    }
  </style>
```

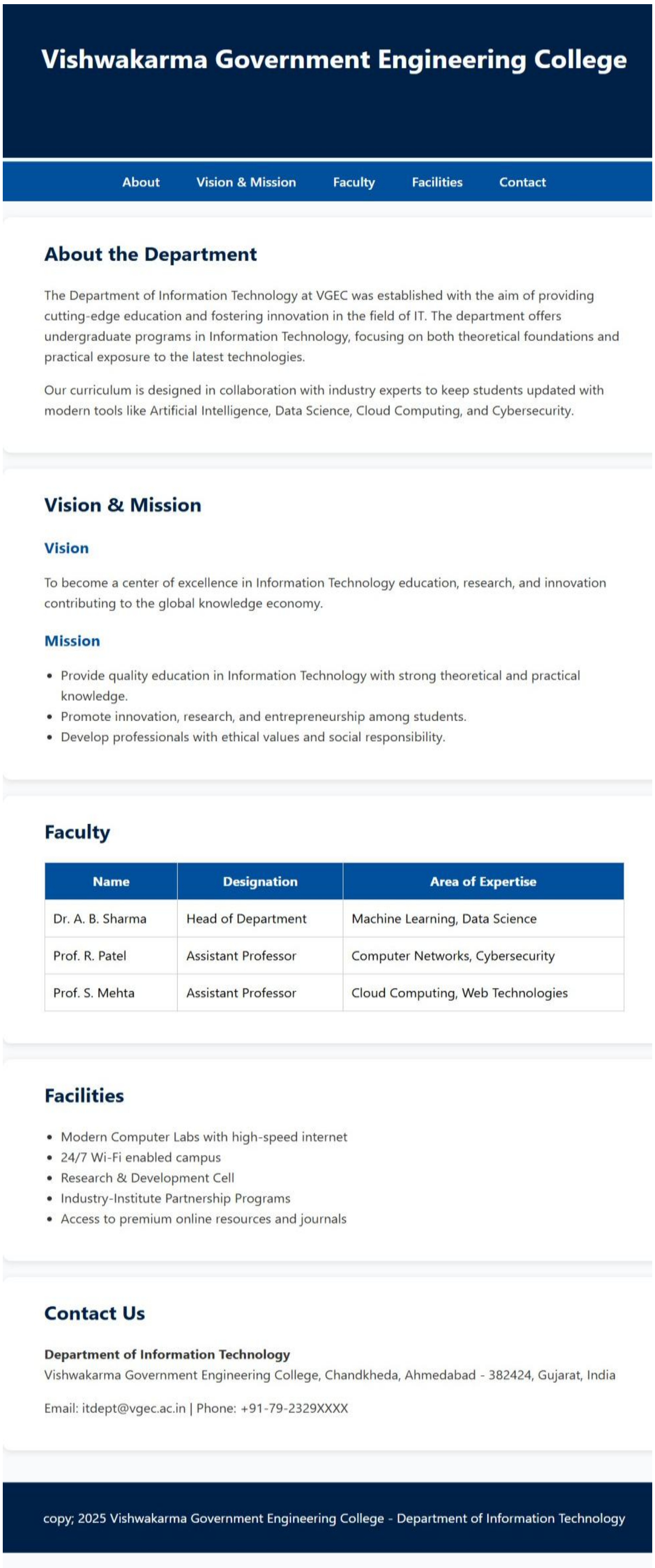
```
</head>
<body>
  <!-- Inline CSS in header border -->
  <header style="border-bottom: 5px solid aliceblue;">
    <h1>Vishwakarma Government Engineering College</h1>
    <h2>Department of Information Technology</h2>
  </header>

  <nav>
    <a href="#about">About</a>
    <a href="#vision-mission">Vision & Mission</a>
    <a href="#faculty">Faculty</a>
    <a href="#facilities">Facilities</a>
    <a href="#contact">Contact</a>
  </nav>

  <section id="about" class="about">
    <h2>About the Department</h2>
    <p>
      The Department of Information Technology at VGEC was established with
      the aim of providing
      cutting-edge education and fostering innovation in the field of IT.
      The department offers
      undergraduate programs in Information Technology, focusing on both theoretical
      foundations
      and practical exposure to the latest technologies.
```

```
    </p>
    <p>
      Our curriculum is designed in collaboration with industry experts to keep
      students updated
      with modern tools like Artificial Intelligence, Data Science, Cloud Computing,
      and Cybersecurity.
    </p>
  </section>

  <section id="vision-mission" class="vision-mission">
    <h2>Vision & Mission</h2>
    <h3>Vision</h3>
    <p>To become a center of excellence in Information Technology education,
    research, and innovation
    contributing to the global knowledge economy.</p>

    <h3>Mission</h3>
    <ul>
      <li>Provide quality education in Information Technology with strong theoretical
       and practical knowledge.</li>
      <li>Promote innovation, research, and entrepreneurship among students.</li>
      <li>Develop professionals with ethical values and social responsibility.</li>
    </ul>
  </section>

  <section id="faculty" class="faculty">
```

```
    <h2>Faculty</h2>

    <table>

      <tr>

        <th>Name</th>

        <th>Designation</th>

        <th>Area of Expertise</th>

      </tr>

      <tr>

        <td>Dr. A. B. Sharma</td>

        <td>Head of Department</td>

        <td>Machine Learning, Data Science</td>

      </tr>

      <tr>

        <td>Prof. R. Patel</td>

        <td>Assistant Professor</td>

        <td>Computer Networks, Cybersecurity</td>

      </tr>

      <tr>

        <td>Prof. S. Mehta</td>

        <td>Assistant Professor</td>

        <td>Cloud Computing, Web Technologies</td>

      </tr>

    </table>

  </section>



  <section id="facilities" class="facilities">
```

```
      <h2>Facilities</h2>
      <ul>
        <li>Modern Computer Labs with high-speed internet</li>
        <li>24/7 Wi-Fi enabled campus</li>
        <li>Research & Development Cell</li>
        <li>Industry-Institute Partnership Programs</li>
        <li>Access to premium online resources and journals</li>
      </ul>
    </section>

    <section id="contact" class="contact">
      <h2>Contact Us</h2>
      <p><strong>Department of Information Technology</strong><br>
        Vishwakarma Government Engineering College, Chandkheda, Ahmedabad -
        382424, Gujarat, India</p>
      <p>Email: itdept@vgec.ac.in | Phone: +91-79-2329XXXX</p>
    </section>

    <footer>
      <p style="color: #fff;">copy; 2025 Vishwakarma Government Engineering College
      - Department of Information Technology</p>
    </footer>
</body>
</html>
```

Output:



Conclusion:

This practical demonstrated the use of **all three types of CSS** together—inline, internal, and external. It improved my understanding of how to separate style concerns and apply professional formatting to a departmental webpage.

Quiz:

1. Explain External CSS.
2. Compare Internal, Inline and External CSS.
3. Which property is used to change the background color?
4. Which property is used to change the text color of the element?

Suggested Reference:

- https://www.geeksforgeeks.org/types-of-css-cascading-style-sheet/

References used by the students:

Rubric wise marks obtained:

Rubrics	1	2	3	Total
Marks				

Experiment No: 11

Develop a java script to display today's date.

Date:

Competency and Practical Skills:

Relevant CO: 4

Objectives:

1. To understand how to write simple java script

Theory:

Javascript

- Javascript is a client side scripting language.
- HTML and CSS for static rendering of a page
- Scripting languges allows content to change dynamically
- Possible to interact with the user beyond what is possible with HTML
- Scripts are programs and can execute on the client side (the one with the browser) or server.
- Running a script on the client saves processing time on the server
- **Types Of Javascript**
    - **Internal Javascript**
        - JavaScript code is placed in the head and body section of an HTML page.
        - **Example**

```
<html>
<head>
<title>Internal JavaScript</title>
<script type="text/javascript">
  document.write("Hello World.!!!");
</script>
</head>
<body>
</body>
</html>
```

-
    - **External javascript**
        - If you want to use the same script on several pages it could be good idea to place the code in separate file, rather than writing it on each.

- JavaScript code are stored in separate external file using the .js extension (Ex: external.js).
- Example :

HTML File : index.html

```
<html>
<head>
<title>External JavaScript</title>
 <script type="text/javascript" src="external.js"></script>
</head>
<body>
</body>
</html>
```

External javascript file : external.js

```
document.write("This is External Javascript Example.!!!");
```

Implementation:

HTML :

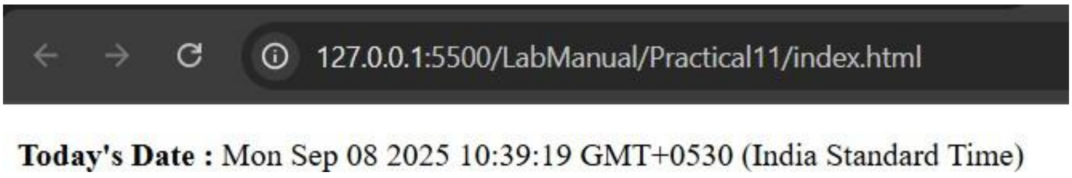
```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Date</title>

</head>

<body>

    <p><b>Today's Date : </b></p>

    <script src="./script.js"></script>

</body>

</html>
```

Develop a java script to display today’s date.

Java Script :

```
let p = document.querySelector('p');

let date = new Date();

p.innerHTML += date;
```

Output :



Conclusion:

By writing a simple JavaScript program to display today's date, I understood how to use the **Date object** in JavaScript. It helped me practice dynamic content rendering on a webpage.

Quiz:

1. What is javascript?
2. Explain internal and external javascript.

Suggested Reference:

- https://www.w3schools.com/JSREF/jsref_obj_date.asp

References used by the students:

Rubric wise marks obtained:

Rubrics	1	2	3	Total
Marks				

Experiment No: 12

Develop simple calculator for addition, subtraction, multiplication and division operation using java script.

Date:

Competency and Practical Skills:

Relevant CO: 4

Objectives:

1. To understand the use of mathematical operators in javascript.
2. To understand the use of document object model.
3. To understand javascript event handling.

Theory:

Javascript Syntax

How to create and use variables?

```
var x,y,z;

x=5;

y=5

z=x+y;

document.write("total is : "+z)
```

The HTML DOM (Document Object Model)

- When a web page is loaded, the browser creates a Document Object Model of the page.
- The HTML DOM model is constructed as a tree of Objects:
- Using DOM Javascript can
    - change all the HTML elements in the page
    - change all the HTML attributes in the page
    - change all the CSS styles in the page
    - remove existing HTML elements and attributes
    - add new HTML elements and attributes
    - react to all existing HTML events in the page

- create new HTML events in the page

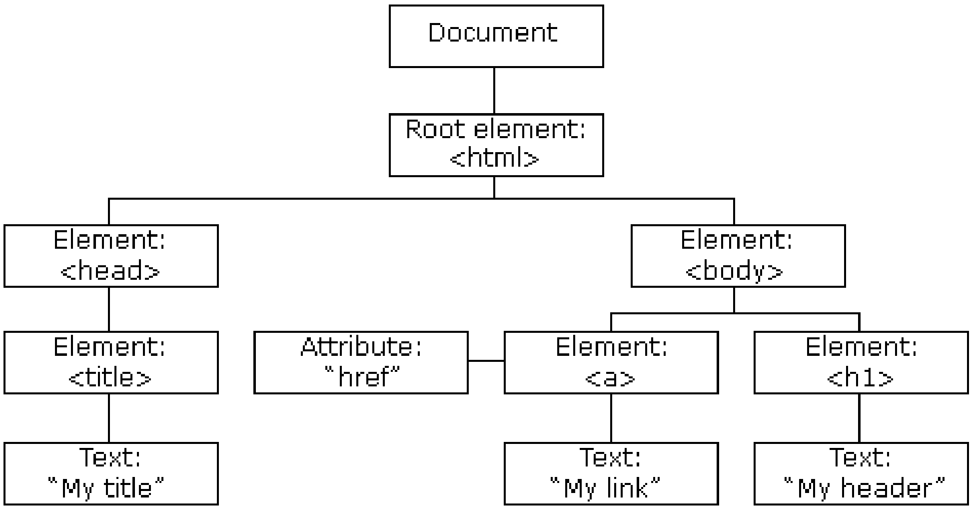


Figure 3 Document Object Model

DOM Examples

Example 1 : following example changes the content of <p> element

```
<html>
<body>

<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML = "Hello World!";
</script>

</body>
</html>
```

Here getElementById is a method, while innerHTML is a property.

Example 2: Validate Numeric Input

```
<!DOCTYPE html>
<html>
<body>

<h2>Number Validation</h2>

<p>Enter a number between 1 and 10:</p>

<input id="numb">

<button type="button" onclick="myFunction()">Submit</button>

<p id="demo"></p>

<script>
function myFunction() {
  // Get the value of the input field with id="numb"
  let x = document.getElementById("numb").value;
  // If x is Not a Number or less than one or greater than 10
  let text;
  if (isNaN(x) || x < 1 || x > 10) {
    text = "Input not valid";
  } else {
    text = "Input OK";
  }
  document.getElementById("demo").innerHTML = text;
}
</script>

</body>
</html>
```

Implementation:

Develop simple calculator for addition, subtraction, multiplication and division operation using java script.

HTML :

```
<!DOCTYPE html>

<html lang="en">



<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Calculator</title>
```

```
    <link href="./style.css" rel="stylesheet">
</head>


<body>
    <div class="mainBox">
        <h2>Calculator</h2>
        <div class="display">
            <div class="input"></div>
            <div class="result"></div>
        </div>
        <div class="content">
            <div class="nums">
                <button>7</button>
                <button>8</button>
                <button>9</button>
                <button>/</button>
            </div>
            <div class="nums">
                <button>4</button>
                <button>5</button>
                <button>6</button>
                <button>*</button>
            </div>
            <div class="nums">
                <button>1</button>
                <button>2</button>
```

```
            <button>3</button>
            <button>-</button>
          </div>
          <div class="nums">
            <button>.</button>
            <button>0</button>
            <button>+</button>
            <button>=</button>
          </div>
        </div>
    </div>
    <script src="./script.js"></script>
</body>

</html>
```

JavaScript :

```
const sum = (...args) =>{
    if (args.length === 0) return 0;
    let ans=0;
    for(let arg of args){
        ans+=arg;
    }
    return ans;
}
const subtract = (...args) => {
```

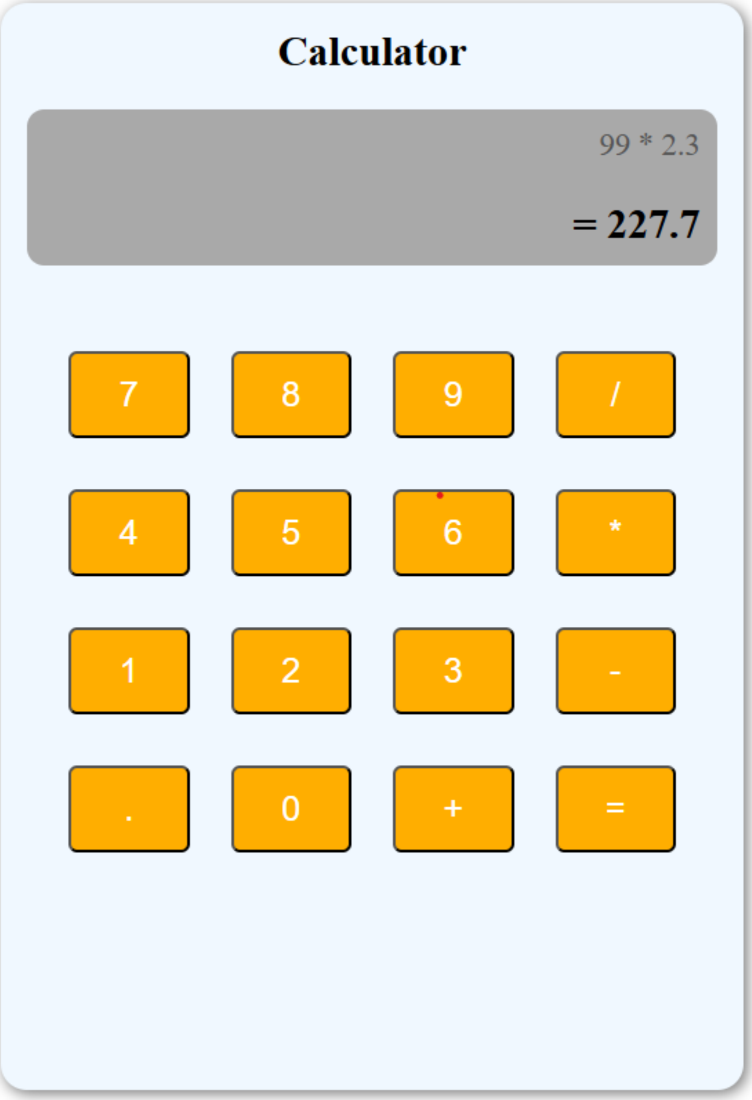
```
    if (args.length === 0) return 0;


    let ans = args[0];
    for (let i = 1; i < args.length; i++) {
        ans -= args[i];
    }
    return ans;
}


const mul = (...args) =>{
    if (args.length === 0) return 0;
    let ans=1;
    for(let arg of args){
        ans*=arg;
    }
    return ans;
}
const div = (...args) =>{
    if (args.length === 0) return 0;
    let ans=args[0];
    for(let i=1; i<args.length;i++){
        if (args[i] === 0) return Infinity;
        ans/=args[i];
    }
    return ans;
}
```

```
const inputDisplay = document.querySelector(".input");
const resultDisplay = document.querySelector(".result");
const buttons = document.querySelectorAll("button");

let currentInput = "";  // ongoing input
let expression = "";   // full expression
let numbers = [];
let operator = null;

buttons.forEach(button => {
  button.addEventListener("click", () => {
    const value = button.innerText;

    if (!isNaN(value) || value === ".") {
      // number or decimal
      currentInput += value;
      inputDisplay.innerText = expression + currentInput;
    }
    else if (["+", "-", "*", "/"].includes(value)) {
      if (currentInput !== "") {
        numbers.push(parseFloat(currentInput));
        expression += currentInput + " " + value + " ";
        currentInput = "";
      } else {
        // replace last operator if no number was typed
```

```
        expression = expression.slice(0, -3) + " " + value + " ";
      }
      operator = value;
      inputDisplay.innerText = expression;
    }
    else if (value === "=") {
      if (currentInput !== "") {
        numbers.push(parseFloat(currentInput));
        expression += currentInput;
      }

      let result;
      switch (operator) {
        case "+": result = sum(...numbers); break;
        case "-": result = subtract(...numbers); break;
        case "*": result = mul(...numbers); break;
        case "/": result = div(...numbers); break;
        default: result = numbers[0];
      }

      inputDisplay.innerText = expression; // show input
      resultDisplay.innerText = "= " + result; // show result

      // reset for next calculation
      numbers = [result];
      currentInput = "";
```

```
        expression = "";
        operator = null;
      }
    });
});
```

Output :



Conclusion:

In this practical, I developed a calculator performing **addition, subtraction, multiplication, and division** using JavaScript. It enhanced my logical thinking and event-handling skills in web development.

Quiz:

1. Explain Document Object Model.

Suggested Reference:

- https://www.w3schools.com/js/js_htmldom.asp
- https://www.w3schools.com/js/js_validation.asp

References used by the students:

Rubric wise marks obtained:

Rubrics	1	2	3	Total
Marks				

Experiment No: 13

Write a java script code to combine and display the information in textbox when the button is clicked use registration page that you created in Practical No.5.

Date:

Competency and Practical Skills:

Relevant CO: 4

Objectives:

1. To understand the use of DOM for getting values from Form Controls.
2. To understand event handling with javascript

Theory:

What is an Event ?

- JavaScript's interaction with HTML is handled through events that occur when the user or the browser manipulates a page.
- When the page loads, it is called an event. When the user clicks a button, that click too is an event. Other examples include events like pressing any key, closing a window, resizing a window, etc.
- Developers can use these events to execute JavaScript coded responses, which cause buttons to close windows, messages to be displayed to users, data to be validated, and virtually any other type of response imaginable.
- Events are a part of the Document Object Model (DOM) Level 3 and every HTML element contains a set of events which can trigger JavaScript Code.

Here is a list of some common HTML events:

Event	Description
onchange	An HTML element has been changed
onclick	The user clicks an HTML element
onmouseover	The user moves the mouse over an HTML element
onmouseout	The user moves the mouse away from an HTML element
onkeydown	The user pushes a keyboard key
onload	The browser has finished loading the page

-

Example : the following javascript example demonstrate how to fetch value from textbox and display using alert()

	``<!DOCTYPE html>
	<html>
	<head>
	
	  <title>Java Script Demo</title>
	  <script>
	    function showData()
	    {
	      var uname,email;
	      uname = document.forms["myform"]["username"].value;
	      email = document.forms["myform"]["email"].value;
	      alert("you entered name:"+uname+" email:"+email);
	    }
	  </script>
	</head>
	<body>
	
	<form name="myform">
	    UserName : <input type="text" name="username"/>
	
	    Password : <input type="email" name="email"/>
	
	    <input type="button" value="display" onclick="showData()" />
	</form>
	</body>
	</html>```	

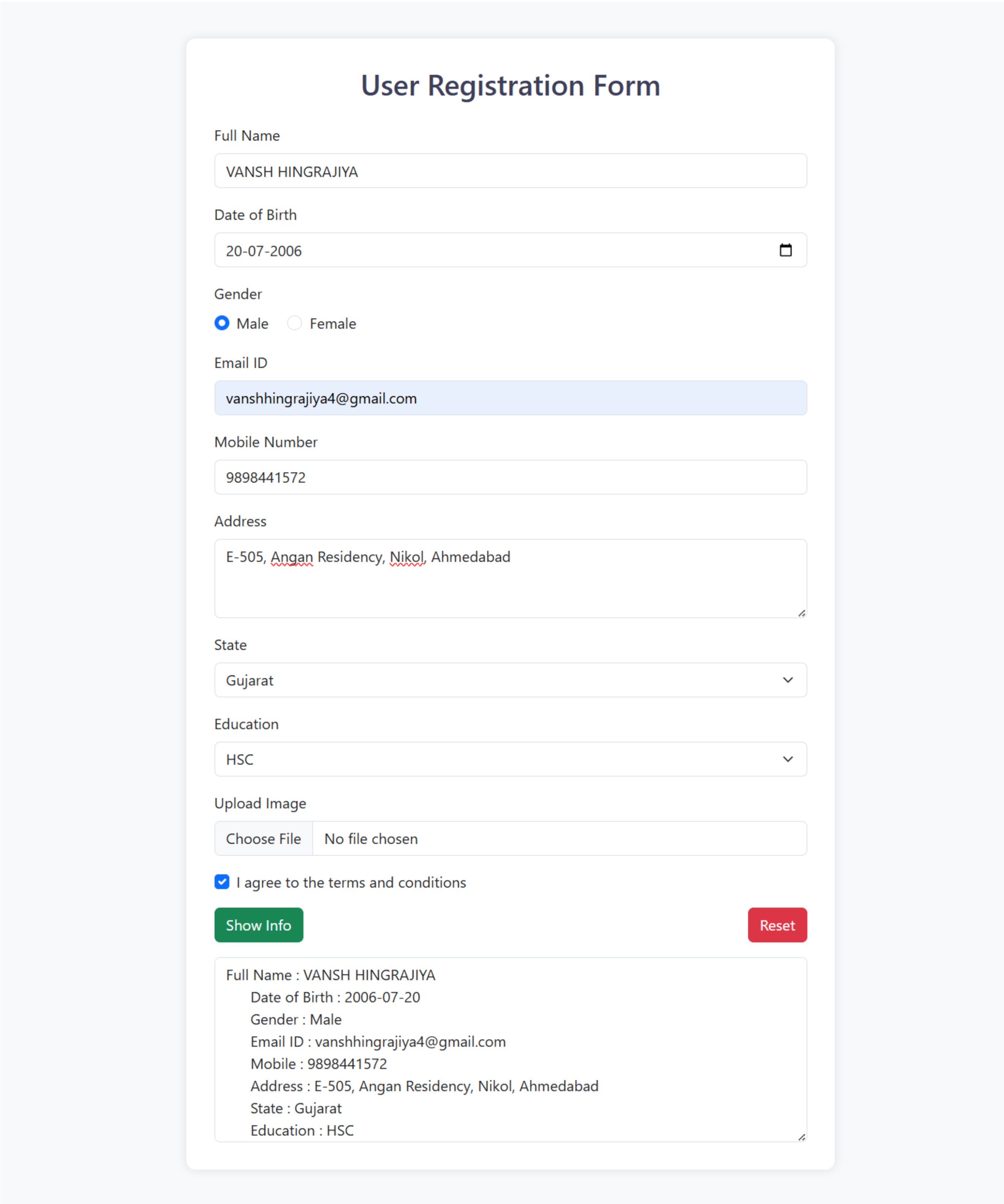
Implementation:

Write a java script code to combine and display the information in textbox when the button is clicked use registration page that you created in Practical No.5.

```
<script>

  function displayInfo() {

    const name = document.getElementById("name").value;

    const dob = document.getElementById("dob").value;

    const gender =

document.querySelector('input[name="gender"]:checked')?.value ||

"Not selected";

    const email = document.getElementById("email").value;

    const mobile = document.getElementById("mobile").value;

    const address = document.getElementById("address").value;

    const state = document.getElementById("state").value;
```

```
const education = document.getElementById("education").value;

    const image =

document.getElementById("image").value.split("\\").pop(); // only

file name


    const info = `

    Full Name : ${name}

    Date of Birth : ${dob}

    Gender : ${gender}

    Email ID : ${email}

    Mobile : ${mobile}

    Address : ${address}

    State : ${state}

    Education : ${education}

    Uploaded Image : ${image}

    `;


    document.getElementById("outputText").value = info.trim();

  }

 </script>
```

Output :


User Registration Form
Full Name
VANSH HINGRAJIYA
Date of Birth
20-07-2006
Gender
Male
Female
Email ID
vanshhingrajiya4@gmail.com
Mobile Number
9898441572
Address
E-505, Angan Residency, Nikol, Ahmedabad
State
Gujarat
Education
HSC
Upload Image
Choose File
No file chosen
I agree to the terms and conditions
Show Info
Reset
Full Name : VANSH HINGRAJIYA
Date of Birth : 2006-07-20
Gender : Male
Email ID : vanshhingrajiya4@gmail.com
Mobile : 9898441572
Address : E-505, Angan Residency, Nikol, Ahmedabad
State : Gujarat
Education : HSC

Conclusion:

This task taught me how to use JavaScript to **collect input values from a registration form and display them** in a combined format. It strengthened my skills in DOM manipulation and user interaction handling.

Quiz:

1. Explain event handling with javascript.

Suggested Reference:

- https://www.w3schools.com/js/js_validation.asp

References used by the students:

Rubric wise marks obtained:

Rubrics	1	2	3	Total
Marks				

Experiment No: 14

Use JavaScript to Implement validation in Practical No.5.

Date:

Competency and Practical Skills:

Relevant CO: 4

Objectives:

1. To understand validation using javascript.

Theory:

Javascript can be used for HTML form validation

Following example demonstrate form validation using javascript

```
<!DOCTYPE html>
<html>
<head>
<script>
function validateForm() {
  let x = document.forms["myForm"]["fname"].value;
  if (x == "") {
    alert("Name must be filled out");
    return false;
  }
}
</script>
</head>
<body>

<h2>JavaScript Validation</h2>

<form name="myForm" action="/action_page.php" onsubmit="return
validateForm()" method="post">
  Name: <input type="text" name="fname">
  <input type="submit" value="Submit">
</form>

</body>
</html>
```

Implementation:

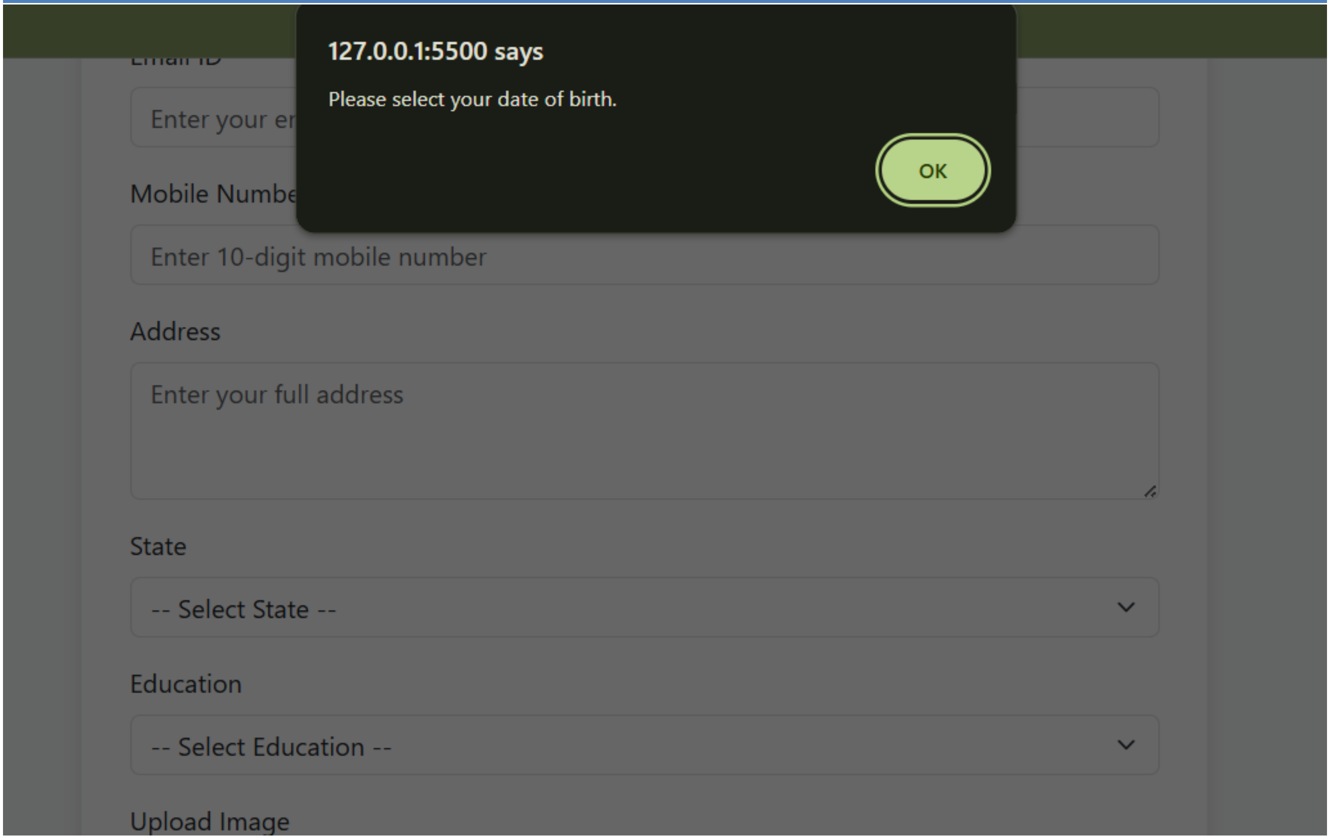
Use JavaScript to Implement validation in Practical No.5

JavaScript :

```
function validateForm() {
  let fname = document.getElementById("fname").value.trim();
  let lname = document.getElementById("lname").value.trim();
  let gender = document.querySelector("input[name='gender']:checked")?.value || "";
  const email = document.getElementById("email").value.trim();
  const birthdate = document.getElementById("birthdate").value;
  const mobile = document.getElementById("mobilenum").value.trim();
  const description = document.getElementById("des").value.trim();

  if (fname === "") {
    alert("First Name is required");
    return false;
  }
  if (lname === "") {
    alert("Last Name is required");
    return false;
  }
  if (gender === "") {
    alert("Please select your Gender");
    return false;
  }
  if (email === "") {
    alert("Email is required");
    return false;
  }
```

```
    let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!emailPattern.test(email)) {
        alert("Enter a valid Email");
        return false;
    }
    if (birthdate === "") {
        alert("Birth Date is required");
        return false;
    }
    if (mobile === "") {
        alert("Mobile Number is required");
        return false;
    }
    if (!/^\d{10}$/.test(mobile)) {
        alert("Mobile Number must be 10 digits");
        return false;
    }
    if (description === "") {
        alert("Description is required");
        return false;
    }
    return true;
}
```

Output :



Conclusion:

By implementing **JavaScript-based form validation**, I learned how to check user inputs like email, mobile number, and required fields. This practical highlighted the importance of validation for data accuracy and user experience.

Quiz:

1. Explain javascript form validation.

Suggested Reference:

- https://www.w3schools.com/js/js_validation.asp

References used by the students:

Rubric wise marks obtained:

Rubrics	1	2	3	Total
Marks				